

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Classified Practice

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.

Classified Practice

Laws of Exponents and Roots

Q003 Written

Find: (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q004 Written

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q005 Written

Find the nine values: 3^0 , 11^0 , $(2/3)^0$, 4^{-3} , 5^{-3} , 6^{-2} , $(1/3)^{-3}$, $(3/5)^{-2}$, $(0.04)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Classified Practice

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3a^{-4} \div (2a)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2).$

Q009 Written**Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.**

Classified Practice

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[3]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Q011 Written**Find** $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice

Laws of Exponents and Roots

Q012 **Written**

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, **and** $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Classified Practice

Laws of Exponents and Roots

Q014 **Written**

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Classified Practice

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Classified Practice

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Classified Practice

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

Classified Practice

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{1/3}]^{1/2}$, $[(1/8)^{1/2}]^{1/3}$, $[(1/16)^{1/4}]^{1/2}$

Classified Practice

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\frac{1000}{27}\right)^{-2/3}\right]^{7/15}$.

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

Classified Practice

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Classified Practice

Laws of Exponents and Roots

Q026 Written

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six **given** expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[5]{a}$ and mixed a,b radicals.

Classified Practice

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Q029 Written

Assess the statement $\sqrt{a^2} = a$ for all real a , using positive and negative examples.

Classified Practice

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Choose the correct option.**

A 8

B $\frac{1}{8}$ C -8 D $\frac{1}{6}$

Q031 MCQ**When multiplying equal bases, we:****Choose the correct option.**

- A subtract exponents
- B add exponents
- C multiply bases only
- D take reciprocals

Classified Practice

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 MCQ**A negative exponent means:****Choose the correct option.**

- A multiply by -1
- B take the reciprocal power
- C set the base to zero
- D add one

Classified Practice

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Choose the correct option.**

- A 25
- B 400
- C 625
- D 1600

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Choose the correct option.**

A -2

B -1

C 2

D 6

Classified Practice

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Choose the correct option.**

A a^7b^6

B $1/(a^7b^6)$

C $a^{-7}b^{-6}$

D $1/(a^6b^7)$

Q037 MCQ**The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:****Choose the correct option.****A** $1/3$ **B** 1**C** 3**D** 9

Classified Practice

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:

Choose the correct option.

A 2

B 4

C 8

D 16

Q039 MCQ $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Choose the correct option.**

A 4

B 5

C 6

D 9

Classified Practice

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Choose the correct option.**A $1/2$ B $1/4$

C 2

D 4

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

- A $\sqrt[3]{2}$
 - B $2\sqrt[3]{2}$
 - C $3\sqrt[3]{2}$
 - D 0
-
-
-

Classified Practice

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 MCQ**When combining roots with the same index, multiply:****Choose the correct option.**

- A the indices
- B the radicands
- C the exponents only
- D the signs only

Classified Practice

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Choose the correct option.**

A 0

B 1

C 2

D 4

Q045 MCQ

$\sqrt[5]{2^4}$ in rational-index form is:

Choose the correct option.

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

Classified Practice

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Choose the correct option.**

A 3

B 6

C 9

D 18

Q047 MCQ**A reciprocal square root is written as:****Choose the correct option.**

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^2

Classified Practice

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Choose the correct option.**

A 27

B $1/9$ C $1/27$ D $1/81$

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

- A $3/4$
- B $4/3$
- C $-3/4$
- D 3

Classified Practice

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

Choose the correct option.

A $4/9$

B $9/4$

C $16/9$

D $27/64$

Q051 MCQ**The 6^{-3} challenge simplifies to:****Choose the correct option.**A $3/10$ B $10/3$ C $100/9$ D $9/100$

Classified Practice

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3} 2^{1/2} \div 2^{7/6}$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Q053 MCQ $4^{5/6} 2^{4/3}$ equals:**Choose the correct option.****A** 4**B** 6**C** 8**D** 16

Classified Practice

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

Choose the correct option.

A $a^{1/3}$ B $a^{2/3}$ C $a^{5/3}$

D a

Q055 MCQ**The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:****Choose the correct option.****A** 2**B** 3**C** 6**D** 12

Classified Practice

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

- A integer powers
- B combining rational exponents
- C logarithms
- D trigonometry

Q057 MCQ**The printed decimal in the 6-4 challenge is:****Choose the correct option.**

- A 0.04
- B 0.004
- C 0.4
- D 4

Classified Practice

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:

Choose the correct option.

- A a always
- B $-a$ always
- C $|a|$
- D 0

Quiz MCQ 1 Concept Quiz · MCQ

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

- A same-base multiplication
- B power of a power
- C reciprocal law
- D root product

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

- A 2
- B 5
- C 10
- D 125

Quiz MCQ 3 Concept Quiz · MCQ

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512

B $1/64$

C $1/512$

D 64

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Choose the correct option.

A a

B $-a$

C 0

D a^2

Quiz WRITTEN 5 Concept Quiz · Written**Simplify** $a^{-3}a^7 \div a^2$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 Concept Quiz · Written**Evaluate** $125^{-2/3}$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Classified Practice

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Q002 Written**Draw $y = 2^x$ and $y = (1/2)^x$.**

Classified Practice

Exponential Functions and Their Applications

Q003 Written

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Q004 Written**Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.**

Classified Practice

Exponential Functions and Their Applications

Q005 **Written**

Order the given sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Q006 Written

Order the three displayed Try sets: $5^0, 5^{-4}, 5^{3/2}$; $(1/3)^7, (1/3)^2, (1/3)^{-5}$; **and** $3^0, 3^2, 3^{-2}$.

Classified Practice

Exponential Functions and Their Applications

Q007 Written

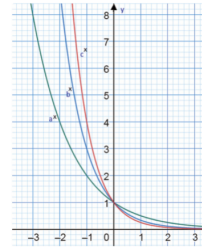
Order the given sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Classified Practice

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c .



Classified Practice

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

Classified Practice

Exponential Functions and Their Applications

Q011 Written

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Classified Practice

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Classified Practice

Exponential Functions and Their Applications

Q015 **Written**

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Classified Practice

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Classified Practice

Exponential Functions and Their Applications

Q019 Written

Solve the three given extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

Q020 Written**For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.**

Classified Practice

Exponential Functions and Their Applications

Q021 Written

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Q022 MCQ

Which base gives a strictly decreasing exponential graph?**Choose the correct option.**

A 2

B 3

C $\frac{1}{2}$ D π

Classified Practice

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

Q024 MCQ

The range of $y = 5^x$ is:**Choose the correct option.**A $\mathbb{R}$ B $[0, \infty)$ C $(0, \infty)$ D $(-\infty, 0)$

Classified Practice

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?

Choose the correct option.

A $(0, \pi)$

B $(1, \pi)$

C $(2, \pi)$

D $(-1, \pi)$

Q026 MCQ**For $0 < a < 1$ and $r < s$, which is true?****Choose the correct option.**

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

Classified Practice

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Choose the correct option.**

- A add exponents
- B make the bases equal
- C take logarithms immediately
- D ignore the base

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Choose the correct option.**A 3^{-2} B 3^0 C 3^2

D They are equal

Classified Practice

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:**Choose the correct option.**A $2^{3/2}$ B 2^2 C 2^3 D 4^3

Q030 MCQ**Solving $4^{x-1} = 8^{x+3}$ gives:****Choose the correct option.**

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

Classified Practice

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Choose the correct option.**

- A the same exponent order
- B a reversed exponent order
- C no base conversion
- D squaring both sides

Q032 MCQ

The solution of $3^x = 81$ is:**Choose the correct option.**

A 2

B 3

C 4

D 5

Classified Practice

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Choose the correct option.

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

Classified Practice

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Choose the correct option.**

A $(3 - \sqrt{73})/2$

B $(3 + \sqrt{73})/2$

C 8

D 16

Q036 MCQ

The challenge quadratic in t is:**Choose the correct option.**

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

Classified Practice

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Choose the correct option.**

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

Q038 MCQ**For extrema, the first required range is the range of:****Choose the correct option.****A** x **B** $t = a^x$ **C** a **D** y only

Classified Practice

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Choose the correct option.**

A -8

B -4

C 0

D 4

Q040 MCQ**The C06-K02 challenge has minimum value:****Choose the correct option.****A** 9 at $x = 0$ **B** 9 at $x = 2$ **C** 27 at $x = 0$ **D** 27 at $x = 2$

Classified Practice

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Choose the correct option.**

A 0

B 1

C 2

D It has no minimum

Q042 MCQ**At which value of x does $a^x + a^{-x}$ attain its minimum?****Choose the correct option.****A** $x = -1$ **B** $x = 0$ **C** $x = 1$ **D** It has no stationary point

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Choose the correct option.

- A range $(0, \infty)$
- B range all reals
- C vertical asymptote
- D decreasing for every a

Quiz MCQ 2 Concept Quiz · MCQ

For base 0.5, increasing the exponent makes the value:

Choose the correct option.

- A larger
- B smaller
- C unchanged
- D zero

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The positive-substitution condition is:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D t can be any real

Quiz MCQ 4 Concept Quiz · MCQ

For $y = 2(t - 1)^2 + 9$, the minimum is:

Choose the correct option.

A 0

B 1

C 9

D 11

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 5 **Concept Quiz · Written**

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Quiz WRITTEN 6 Concept Quiz · Written**Order** 2^{-3} , 2^0 , $2^{5/2}$.

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

Quiz WRITTEN 8 Concept Quiz · Written

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote. **Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively. **Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$. **Q027** B **Q028** C **Q029** B **Q030** A **Q031** B **Q032** C **Q033** B **Q034** C **Q035** B **Q036** B **Q037** C **Q038** B **Q039** B **Q040** A **Q041** C **Q042** B **Quiz MCQ 1** A **Quiz MCQ 2** B **Quiz MCQ 3** C **Quiz MCQ 4** C **Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$ **Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$ **Quiz WRITTEN 7** $x = 0$ or $x = 2$ **Quiz WRITTEN 8** $2atx = 0$
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$. **Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x . **Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$. **Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$. **Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations. **Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$. **Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$. **Q022** C **Q023** B **Q024** C **Q025** B **Q026** B
- Q003** Bases $4, 10, 3/2$: strictly increasing. Bases $2/3, 1/4, 1/10$: strictly decreasing. Every graph has $D = \mathbb{R}, R = (0, \infty)$, and asymptote $y = 0$. **Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control. **Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours. **Q013** First: let $t = 2^x > 0, t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$. **Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$. **Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q004** Key points $(-1, 1/\pi), (0, 1), (1, \pi), (2, \pi^2)$; the restricted graph is increasing. **Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$. **Q006** $5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2$. **Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$. **Q008** Rewrite every term as a power of 2 : $2^{5/2}, 2^2, 2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Classified Practice

Introduction to Logarithms and Their Properties

Q001 **Written**

Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Classified Practice

Introduction to Logarithms and Their Properties

Q003 **Written**

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Q004 Written

If $\log_5 25 = x$, **evaluate** $\log_{1/2} x$ **without a calculator**.

Classified Practice

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1$, $\log_5 5$, $\log_2 8$, $\log_2 16$, $\log_5(1/5)$, $\log_7(49^{3/2})$.

Q006 Written

Find the six Try values $\log_5 1$, $\log_3 3$, $\log_4 4$, $\log_3 27$, $\log_2(1/4)$, $\log_5(1/125)$.

Classified Practice

Introduction to Logarithms and Their Properties

Q007 **Written****Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.**

Q008 Written**Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.**

Classified Practice

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4 \log_5 3 - 2 \log_5 15 - \log_5 45$.

Classified Practice

Introduction to Logarithms and Their Properties

Q011 Written

Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.

Classified Practice

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24$, $\log_{10}(27/4)$, $\log_{10} 45$, $\log_{10}(24/75)$ in terms of a, b .

Q014 Written

Given $\log_{10} 2 = a$, $\log_{10} 3 = b$, express the four Try arguments 72 , $3^8/10$, 15 , 12 in terms of a , b .

Classified Practice

Introduction to Logarithms and Their Properties

Q015 **Written**

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Q016 Written

If $\log_c 3 = x$, $\log_c 2 = y$, **write** $\log_c 12$, $\log_c(3/8)$, $\log_c \sqrt{6}$, $\log_2 3$, $\log_{3^b}(2^b c)$ **in terms of** x, y .

Classified Practice

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Classified Practice

Introduction to Logarithms and Their Properties

Q019 **Written**

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Q020 Written**Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.**

Classified Practice

Introduction to Logarithms and Their Properties

Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Q022 Written**Solve the four Try products of logarithms after making the bases the same.**

Classified Practice

Introduction to Logarithms and Their Properties

Q023 **Written****Simplify all 15 Exercise products and sums of logarithms by common-base conversion.**

Q024 Written**Simplify** $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ **for** $x > 1$.

Classified Practice

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

Classified Practice

Introduction to Logarithms and Their Properties

Q027 Written

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.

Q028 Written**Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.**

Classified Practice

Introduction to Logarithms and Their Properties

Q029 Written

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Q030 Written

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2\log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Classified Practice**Introduction to Logarithms and Their Properties****Q031** Written**Order the four Try sets of logarithms using monotonicity and change of base.**

Q032 Written**Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.**

Classified Practice

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Q034 MCQ $\log_2 32$ equals:

Choose the correct option.

A 2

B 4

C 5

D 32

Classified Practice

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:

Choose the correct option.

A -2

B -1

C 1

D 2

Q036 MCQ $\log_5(1/125)$ equals:

Choose the correct option.

A -3

B -2

C 2

D 3

Classified Practice

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:

Choose the correct option.

A 1

B 2

C $3/2$ D $7/2$

Q038 MCQ

$\log_a M + \log_a N$ equals:

Choose the correct option.

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

Classified Practice

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Choose the correct option.**

A $\log_5(2 + 3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

Q040 MCQ**If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:****Choose the correct option.**

- A $a + b$
 - B $2a + 3b$
 - C $3a + b$
 - D $3a + 2b$
-
-
-

Classified Practice

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Choose the correct option.

- A $x + y$
- B $(x + y)/2$
- C $2x + y$
- D $x + 2y$

Q042 MCQ $\log_{16} 64$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Choose the correct option.**

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

Q044 MCQ $\log_4 16 \cdot \log_9 3$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Choose the correct option.**

A 6

B 8

C 12

D 24

Q046 MCQ

The domain of $y = \log_a x$ is:**Choose the correct option.**A all real x B $x \geq 0$ C $x > 0$ D $x \neq 0$

Classified Practice

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Choose the correct option.

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

Q048 MCQ**For $a > 1$ and $M < N$, which is true?****Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal**D** Cannot compare

Classified Practice

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

Q050 MCQ**The vertical asymptote of $y = \log_a x$ is:****Choose the correct option.**

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

Classified Practice

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Choose the correct option.

- A $\log_a M^{-2}$
- B $\log_a(-2M)$
- C $\log_a(M - 2)$
- D $2 \log_a M$

Q052 MCQ

The base of $\log_a M$ is:**Choose the correct option.**A M B a C p D $1/a$

Classified Practice

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Choose the correct option.**A a B M C p D 0

Q054 MCQ $\log_2(1/8)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:

Choose the correct option.

A 0

B 1

C 2

D 4

Q056 MCQ $\log_3 1$ equals:**Choose the correct option.****A** -1**B** 0**C** 1**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:

Choose the correct option.

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

Q058 MCQ **$2 \log_3 5$ equals:****Choose the correct option.****A** $\log_3 10$ **B** $\log_3 25$ **C** $\log_3 7$ **D** $\log_3(5/2)$

Classified Practice

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Choose the correct option.

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Choose the correct option.

A a

B $1 - a$

C $1 + a$

D $-a$

Classified Practice

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:

Choose the correct option.

A $2/3$ B $3/2$ C $5/3$

D 3

Q062 MCQ $\log_{25} 125$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D $5/2$

Classified Practice

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:

Choose the correct option.

A 0

B 1

C 2

D $\log_a b$

Q064 MCQ

$\log_2 8 \cdot \log_8 2$ equals:

Choose the correct option.

A $1/3$

B 1

C 3

D 8

Classified Practice

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Choose the correct option.**

A 2

B 4

C 8

D 10

Q066 MCQ**The range of $y = \log_a x$ is:****Choose the correct option.****A** $x > 0$ **B** $\mathbb{R}$ **C** $[0, \infty)$ **D** $y > 0$

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Choose the correct option.

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

Quiz MCQ 2 Concept Quiz · MCQ

$\log_2 8 + \log_2 4$ equals:

Choose the correct option.

A 2

B 3

C 5

D 12

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D 4

Quiz MCQ 4 Concept Quiz · MCQ

For $a > 1$, $M < N$ implies:

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written

Find $\log_3(1/27)$.

Quiz WRITTEN 6 Concept Quiz · Written**Simplify** $\log_5 25 + \log_5 5$.

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Quiz WRITTEN 8 Concept Quiz · Written

State the domain and range of $y = \log_a x$.

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4(1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1 A****Quiz MCQ 2 C****Quiz MCQ 3 B****Quiz MCQ 4 B****Quiz WRITTEN 5 -3** **Quiz WRITTEN 6 3****Quiz WRITTEN 7 $3/2$** **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Classified Practice

Applications of Logarithms

Q001 **Written****Solve** $\log_2 x = 4$.

Q002 Written**Solve** $\log_2(2x - 1) = \log_2 3$.

Classified Practice

Applications of Logarithms

Q003 **Written**

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Q004 **Written****Solve** $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ **completely.**

Classified Practice

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.

Classified Practice

Applications of Logarithms

Q007 **Written**

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 Written

Solve the given challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Classified Practice

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16 \log_2 x + 3 < 0$.

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

Classified Practice

Applications of Logarithms

Q011 Written

Solve the **given** groups: three equations, three inequalities, and three exact logarithmic values.

Q012 Written**Solve** $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x .

Classified Practice

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Classified Practice

Applications of Logarithms

Q015 **Written**

Solve the three given extrema problems, each with a different base and interval.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Classified Practice

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 Written**Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.**

Classified Practice

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Q020 Written**Solve** $\log_x(x + 2) \leq \log_x(4 - x)$ **for all real** x .

Classified Practice

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Choose the correct option.**A $x < 5$ B $x \geq 5$ C $x > 5$

D all real x

Q022 MCQ**If $\log_2(2x - 1) = \log_2 3$, then $x =$:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Choose the correct option.**

A 1

B 2

C 4

D 8

Q024 MCQ**For same-base logarithms, equality of logs gives:****Choose the correct option.**

- A arguments equal
- B bases equal
- C arguments add
- D no result

Classified Practice

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A $M > N$

B $M < N$

C $M = N$

D $MN < 0$

Q026 MCQ**For base $0 < a < 1$, the inequality direction is:****Choose the correct option.**

- A preserved
- B reversed
- C removed
- D squared

Classified Practice

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Choose the correct option.**A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

Q028 MCQ

The warm-up intersection is:**Choose the correct option.**A $0 < x < 5$ B $5 < x < 6$ C $x > 6$ D $-3 < x < 6$

Classified Practice

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Choose the correct option.**

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ**The inequality $5t^2 - 16t + 3 < 0$ gives:****Choose the correct option.****A** $t < 1/5$ **B** $1/5 < t < 3$ **C** $t > 3$ **D** $0 < t < 1$

Classified Practice

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Choose the correct option.**A $t > 0$ B $t \in \mathbb{R}$ C $t < 0$ D $t = 1$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

Choose the correct option.

A $x < 1/9$

B $1/9 < x < 27$

C $x > 27$

D $0 < x < 3$

Classified Practice

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$

B $t = 1$

C $t = 2$

D $t = 3$

Q034 MCQ

The correct first step for extrema is:**Choose the correct option.**

- A differentiate in x
- B set $t = \log_a x$
- C ignore the interval
- D take $x = 0$

Classified Practice

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Choose the correct option.**A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

Q036 MCQ**The water-tank challenge maximum health rating is:****Choose the correct option.****A** 3**B** 4**C** 5**D** 8

Classified Practice

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Choose the correct option.**

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

Q038 MCQ**If $-9 < \log_{10} N < -8$, the first non-zero digit is at:****Choose the correct option.**

- A 8th
- B 9th
- C 10th
- D 11th

Classified Practice

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Choose the correct option.**

A 2

B e

C 10

D any base

Q040 MCQ**The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:****Choose the correct option.****A** (0,1)**B** (1,4)**C** (0,4)**D** empty

Classified Practice

Applications of Logarithms

Quiz MCQ 1 Concept Quiz · MCQ

Solve $\log_5(x - 1) = 2$.

Choose the correct option.

A $x = 26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 Concept Quiz · MCQ

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.**Choose the correct option.**A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$.

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.

Classified Practice

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 Concept Quiz · Written

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits